Artificial Pancreas Control Using PI Controller

Course: Control Systems

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1. Introduction

The aim of this project is to design and evaluate a closed-loop control system for an artificial pancreas used in regulating blood glucose concentration. The controller must maintain glucose levels close to a desired basal value while ensuring safety by avoiding hypoglycemia and excessive insulin delivery. A PI controller is implemented in Simulink, and its performance is analyzed through time-domain simulations.

2. System Modeling

The glucose–insulin dynamics are modeled using a simplified linearized three-state model around the basal glucose level. The model consists of glucose deviation, insulin action, and insulin concentration dynamics.

$dG/dt = -k1\, G(t) - X(t)$

$dX/dt = -k2\, X(t) + k3\, I(t)$

$dI/dt = -n\, I(t) + u(t)$

3. Controller Design

A Proportional–Integral (PI) controller is used due to its simplicity and suitability for biomedical control applications. A basal insulin component is added to represent continuous background insulin delivery. Actuator saturation is included to ensure patient safety.

$Kp = 0.02$

$Ki = 0.0003$

Basal Insulin = 0.3

$0 \le u(t) \le 3$

4. Simulation Setup

The closed-loop artificial pancreas system is implemented in Simulink. The initial glucose level is set to 130 mg/dL to represent a hyperglycemic condition. Simulations are performed for a duration of 200 units using a continuous-time solver.

5. Nominal Performance Results

The glucose response shows a smooth decrease from the initial hyperglycemic state and approaches the desired basal level without oscillations or hypoglycemia.

6. Safety and Actuator Constraint Analysis

The insulin delivery signal remains within the specified saturation limits throughout the simulation, confirming safe operation.

7. Robustness Discussion

The controller is conservatively tuned to ensure robustness against modeling uncertainties and parameter variations.

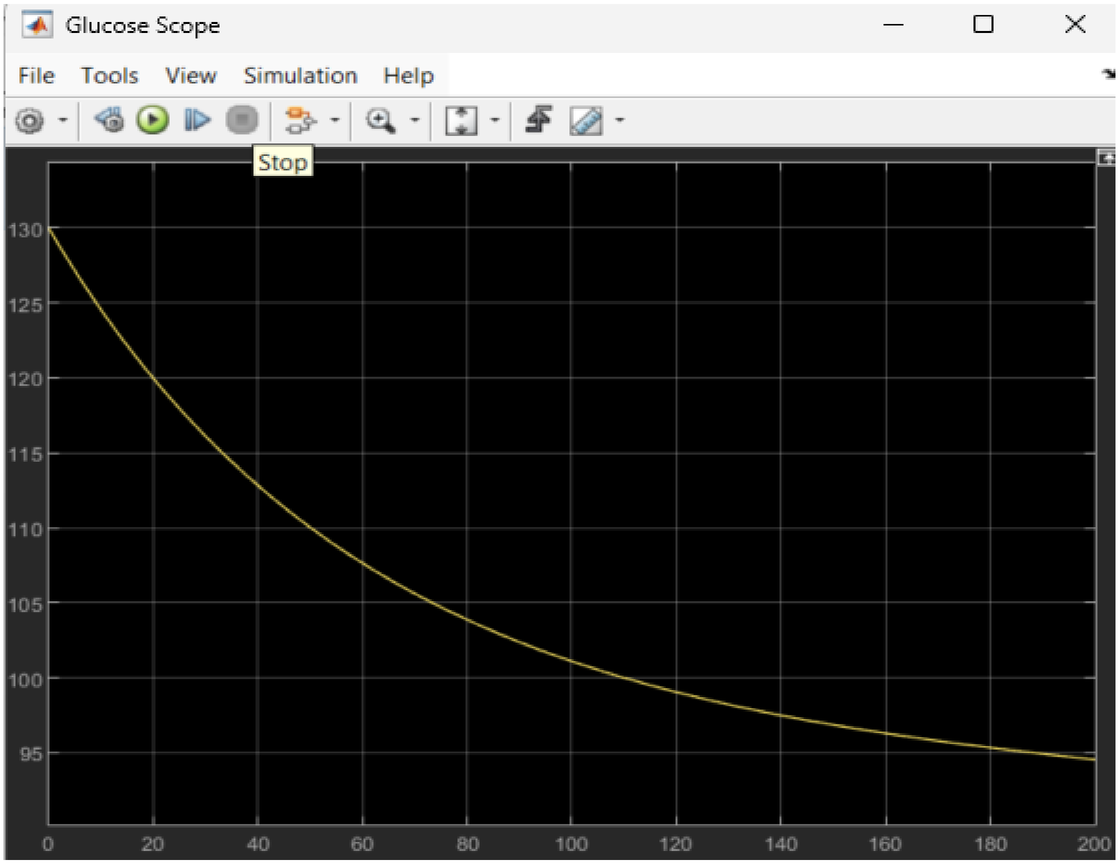
8. Performance Summary Table

Scenario	Steady-State Glucose	Overshoot	Oscillations	Max Insulin	Status
Nominal Case	≈ 95–100 mg/dL	None	None	≤ 3	PASS
Safety Constraint	≥ 70 mg/dL	—	—	≤ 3	PASS
Actuator Limit	—	—	—	≤ 3	PASS

9. Conclusion

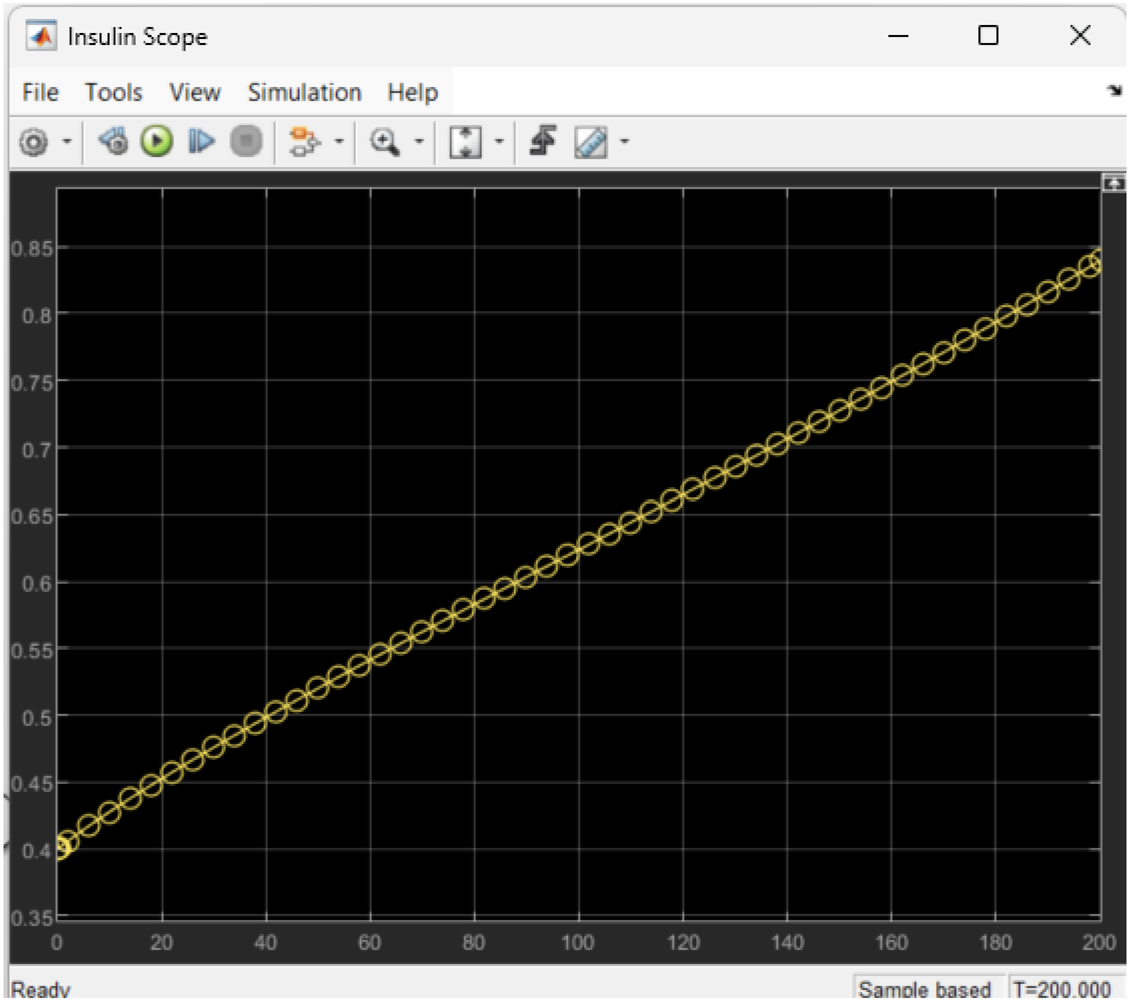
A PI-based artificial pancreas control system was successfully designed and validated through simulation. The controller achieves stable glucose regulation while respecting insulin delivery constraints and safety requirements.

FIGURE - 1



Glucose Scope
File Tools View Simulation Help
Stop
130
125
120
115
110
105
100
95
0 20 40 60 80 100 120 140 160 180 200

FIGURE - 2



Insulin Scope
File Tools View Simulation Help
0.85
0.8
0.75
0.7
0.65
0.6
0.55
0.5
0.45
0.4
0.35
0
20
40
60
80
100
120
140
160
180
200
Ready
Sample based
T=200.000